

Lipschitz Continuity

Reference for ODE existence & uniqueness $\dot{x} = f(t, x)$ and nonlinear control

1 Definition

$f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **Lipschitz** on a set S if $\exists L \geq 0$ such that

$$\|f(x) - f(y)\| \leq L \|x - y\| \quad \forall x, y \in S.$$

The smallest such L is the *Lipschitz constant*. Geometrically L caps the slope of the secant line between *any* two points. For a differentiable f , $L = \sup_S \|f'\|$.

2 The three grades

GL Globally Lipschitz: one L works on *all* of $\mathbb{R}^n$; slope bounded everywhere. e.g. $\sin x$, $\|x\|$, Ax .

LL Locally Lipschitz: every point has a neighborhood on which f is Lipschitz (L may vary), but no single global L . e.g. x^2 , e^x , any C^1 map.

NLL Not locally Lipschitz (at a point): slope unbounded in *every* neighborhood of some x_0 . e.g. $\sqrt{x}$, $x^{1/3}$ at 0.

Hierarchy (all strict):

$$\text{GL} \subset \text{LL} \subset C^0, \quad C^1 \subset \text{LL}.$$

3 Why it matters (Picard–Lindelöf)

For $\dot{x} = f(t, x)$, $x(t_0) = x_0$ with f Lipschitz in x :

- LL** in $x \Rightarrow$ *unique* solution, locally in time.
- GL** in $x \Rightarrow$ unique solution for *all* t (no finite escape).

Failure modes:

- NLL** $\Rightarrow$ uniqueness can break: $\dot{x} = 2\sqrt{x}$, $x(0) = 0$ has both $x \equiv 0$ and $x = t^2$.
- LL** but not **GL** $\Rightarrow$ finite escape: $\dot{x} = x^2$, $x(0) = 1$ gives $x = \frac{1}{1-t}$, blowing up at $t = 1$.

4 How to check — recipe

1. **Differentiable?** C^1 on an open set $\Rightarrow$ locally Lipschitz *automatically*. The local question is then settled; only globality is open.

2. **Bound the derivative**:

$$\text{scalar: } L = \sup_S |f'|, \quad \text{vector: } L = \sup_S \|J\|, \quad J = \frac{\partial f}{\partial x}.$$

3. **Read off** the supremum:

- finite over all $\mathbb{R}^n \rightarrow$ **GL** (that sup *is* L);
- finite on every *bounded* set, ∞ overall $\rightarrow$ **LL**;
- $\|f'\| \rightarrow \infty$ near a point $x_0 \rightarrow$ **NLL** at x_0 .

4. **Not differentiable** (corners: $|x|$, ReLU, sat, min / max): use the *definition* directly, or the algebra below. A Lipschitz f is differentiable a.e. with $L = \text{ess sup} \|\nabla f\|$ — kinks on measure-zero sets don't raise L .

Convexity caveat. The Jacobian bound needs the segment $[x, y] \subseteq S$. On a *non-convex* domain, a bounded J alone does *not* give the global L .

5 Building blocks & traps

- Any norm is 1-Lipschitz** (reverse triangle ineq.): $\|x\| - \|y\| \leq \|x - y\|$. Linear map: $\|Ax - Ay\| \leq \|A\| \|x - y\| \Rightarrow$ **GL**, $L = \|A\|$.
- Norm powers** $\|x\|^p$: $p=1 \rightarrow$ **GL** ($L=1$); $p>1 \rightarrow$ **LL** (gradient $\sim \|x\|^{p-1}$ unbounded); $0 < p < 1 \rightarrow$ **NLL** (cusp at 0).
- Two danger zones**: the *origin* (cusps, $p < 1$) and *infinity* (super-linear growth, $p > 1$). Always check both.
- Composition** inherits **GL** only from the *inner* map: $\sin \circ g$ is **GL** iff g is. $\sin$ is 1-Lipschitz, yet $\sin(x^2)$ is only **LL**.
- Algebra**: sums (constants add), compositions (constants multiply), products of *bounded* Lipschitz functions — all stay Lipschitz.
- Boundedness of f is irrelevant**: $\sin x$ is bounded & **GL**; x is unbounded & **GL**. What matters is the *slope*.

6 Worked example — classifying nine functions

#	function	grade	reason
i	x^2	GL/LL $\rightarrow$ LL	smooth $\Rightarrow$ locally Lipschitz; $f' = 2x$ unbounded $\Rightarrow$ not global.
ii	$ x ^{1/2}$	NLL	$f' \sim \frac{1}{2} x ^{-1/2} \rightarrow \infty$ at 0 (cusp).
iii	$\text{sign}(x) x ^{1/2}$	NLL	difference quotient at 0 is $\sqrt{ x }/ x = 1/\sqrt{ x } \rightarrow \infty$.
iv	$\ x\ _2^2 = x^\top x$	LL	smooth; gradient $2x$ unbounded $\Rightarrow$ not global.
v	$\ x\ _2$	GL	reverse triangle inequality gives $L = 1$; kink at 0 is harmless.
vi	$\ x\ _2^{1/2}$	NLL	behaves like $\sqrt{\ x\ }$; quotient $\rightarrow \infty$ at the origin.
vii	$\ Ax\ _2$	GL	$ \ Ax\ - \ Ay\ \leq \ A(x - y)\ \leq \ A\ \ x - y\ $, so $L = \ A\ $.
viii	$\sin(x^\top Ax)$	LL	smooth; gradient $\cos(\cdot)(A + A^\top)x$ — $\cos$ bounded but $(A + A^\top)x$ grows.
ix	$\sin\ x\ _2$	GL	1-Lipschitz $\circ$ 1-Lipschitz $\Rightarrow L = 1$.

Rule of thumb: C^1 settles *local*; the derivative's growth at infinity settles *global*; a blow-up at a finite point (usually the origin) is the only way to fail *local*.